GM System: Technical Description and Pitch

Contactless Ebola screening, contact monitoring and triage, tied to one patient record. Technical description v1.0, 15 October 2026. GM stands for Gusmao-Mbaja, the designers' names.

1. Executive summary

Kenya has confirmed two cases of Ebola (Bundibugyo virus disease) linked to an imported index case, one patient has died, and the Ministry of Health has declared a formidable epidemic disease and set up a single national command. The outbreak in the Democratic Republic of the Congo driving these cases has passed 9,000 confirmed cases, and no approved vaccine or specific treatment exists for this strain. Kenya's response therefore rests on three things: finding sick people early, monitoring contacts for 21 days, and getting laboratory confirmation back to the right people fast.

The GM System supports all three with two field devices and one platform:

- **Stage 1, the GM Checkpoint:** a touch-free station for airports, border posts and hospital entrances. It measures forehead temperature, asks exposure questions by hand gesture, and gives an immediate green, amber or red referral decision. An experimental electronic-nose module records gas-sensor data alongside, without affecting the decision until it is validated.
- **Stage 2, the GM Wearable:** a sealed wrist monitor issued to contacts in quarantine. It tracks skin temperature, heart rate, SpO2 and movement through the 21-day watch period and alerts a health worker when the trend turns, even if the person is at home.
- **The GM Platform:** one patient record, a command-centre dashboard, rule-based decision support, and standards-based data exchange (HL7 FHIR) with hospitals, national reporting systems and reference laboratories. Laboratory results return through the same patient ID, which closes the loop without a third device.

The GM System is a screening, monitoring and triage aid. It does not diagnose Ebola; confirmation stays with the national laboratories.

Status. The design is complete for both stages, with every key component checked against manufacturer datasheets. Nothing has been built or validated yet. This document sets out the technical design, the validation and approval plan, and the support needed to run a pilot.

2. Situation in Kenya

As reported between 6 and 15 October 2026. Figures change quickly, so confirm them with the Ministry of Health and WHO before use.

- **First case:** a 40-year-old Kenyan who had lived in the DRC for seven years arrived in Nairobi on 3 October via Uganda, died on 5 October, and was confirmed on 6 October.
- **Second case:** a 39-year-old close relative, already in quarantine under daily monitoring, developed fever, fatigue and joint pains on 13 October. The National Virology Reference Laboratory and the KEMRI-CDC facility confirmed the infection on 14 October. Officials said early detection improved his chances of recovery and showed that contact tracing is working.
- **Contacts:** 128 identified for the first case, 108 placed in quarantine, 113 traced by 13 October, and five contacts outside Kenya notified to their authorities. The Ministry reports no evidence of wider community spread.
- **Legal response:** the Health Cabinet Secretary declared Ebola a formidable epidemic disease on 12 October. A single national command coordinates under the Public Health (Declaration of Formidable Epidemic Disease) Order, 2026 and the Public Health (Ebola Disease) Rules, 2026.
- **Borders:** screening has been tightened at borders and airports. WHO and the UN advise against travel and trade restrictions, so screening has to work without closing borders.
- **Regional outbreak:** Bundibugyo virus disease in the DRC was declared in May 2026 and has been a public health emergency of international concern since 16 May, with over 9,000 confirmed cases and over 4,300 deaths.

The second case shows what works: a contact who is already under monitoring is found early. The difficulty is doing that at scale. By simple arithmetic, 128 contacts followed for 21 days is 2,688 contact-days of daily follow-up for a single index case.

Why Ebola control depends on contact handling. Ebola spreads through direct contact with the blood or body fluids of a person who is sick or has died of it, and symptoms can appear 2 to 21 days after exposure. It is not spread by casual contact with intact skin, but contaminated surfaces and samples are the real hazard. Early identification, isolation and contact monitoring are the main tools of control.

3. Operational gaps the system addresses

- **Slow, contact-heavy entry screening.** Screening that depends on staff handling thermometers and paper forms close to possibly infectious travellers is slow and exposes the people doing it.
- **Blind follow-up between checks.** Contact follow-up by repeated manual checks is labour-intensive, and nothing is known about a contact between visits or calls.
- **Fragmented records.** Screening results, monitoring data and laboratory results sit in separate places, so the command centre decides on partial information.
- **Late or lost laboratory results.** Delays hold up isolation decisions and the release of contacts who test negative.

4. What the GM System is

Component	Where it is used	What it does	Output
GM Checkpoint (Stage 1)	Airports, border posts, hospital entrances	Touch-free temperature and exposure screening, identification and enrollment	Green, amber or red decision; record to the platform
GM Wearable (Stage 2)	Contacts in quarantine or home monitoring	Continuous trend monitoring with local and remote alerts	Trend alerts, heartbeats and readings to the patient record
GM Platform	Command centre, hospitals, laboratories	Patient record, dashboard, decision support, data exchange, laboratory results by patient ID	Alerts, case status, FHIR exchange

Laboratory results are not a device stage. Samples are collected and tested by trained staff in the national laboratories under the health authority's protocols. The platform receives the result against the patient ID and specimen number and updates the person's status.

Design principles

- **Touch-safe by design:** remove touch points first, then make what remains easy to disinfect.
- **Screen, don't diagnose:** every output is a referral decision with an audit trail.
- **Never fail silently:** heartbeats, retries, offline buffering and local alerts mean a dead battery or a lost link is itself an alert.
- **Verify before trusting:** parts are chosen against datasheets and every accuracy claim is validated against reference instruments.
- **Privacy by default:** pseudonymous IDs, minimum data, encryption, informed consent.
- **Standards-based:** HL7 FHIR for exchange, so the platform plugs into national systems instead of replacing them.
- **Build and validate one stage at a time.**

5. System architecture

```
[Wearable] --BLE--> [Phone app] --internet/SMS--> [Platform]
[Wearable] --cellular (fallback, later)--------> [Platform]
[Checkpoint] --Wi-Fi--> [Site gateway] --LTE/GSM or internet--> [Platform]
[Laboratory] --FHIR / HL7 results by patient ID--> [Platform]
[Platform] <--FHIR / DHIS2 exchange--> [Hospital systems, Kenya HIE, national
reporting]
```

The patient ID lifecycle

One pseudonymous patient ID ties every stage together.

1. The first tap of an NFC wristband at a checkpoint creates a provisional record keyed to the tag ID, and screening runs immediately.
2. Amber or red cases and monitored contacts are enrolled with consent at the operator desk, and a wearable is bound to the same ID.
3. Samples collected under health authority protocols carry the same ID on the specimen label.
4. Laboratory results post back to the platform against that ID and specimen number, updating status and alerts.

6. Stage 1: GM Checkpoint

Purpose. Screen each person for fever and exposure risk without touch, give an immediate visible and audible result, and issue or confirm the patient ID.

Flow

1. The person stands on a floor marker. A time-of-flight sensor confirms they are inside the validated measuring window (8 to 15 cm).
2. An NFC wristband tap identifies them. An unknown tag creates a provisional record.
3. After the position is stable and the sensor bus has been quiet for about a second, the medical-grade infrared sensor takes a short burst of readings, which are averaged.
4. Spoken prompts and on-screen text ask three to five yes/no exposure and symptom questions, answered by hand gesture.
5. A rules engine sets green, amber or red. The result is shown and announced.
6. The record goes to the site gateway and is buffered locally if the link is down.
7. A use counter increments, and the unit locks until an operator confirms cleaning with a badge.

Hardware and the datasheet facts that shape it

Block	Part	Datasheet facts that shape the design
Controller	ESP32-S3-WROOM-1-N16R2 module	16 MB flash, 2 MB quad PSRAM, -40 to 85 degC, 3.0 to 3.6 V, Wi-Fi and BLE. The N16R8 was rejected because its octal PSRAM limits it to 65 degC ambient
Thermometer	MLX90614ESF-DCI (medical grade, 3 V, 5 degree field of view)	+/-0.2 degC accuracy within a limited window (sensor package 16 to 40 degC, object 22 to 40 degC); calibrated at 3.0 V +/-50 mV with about 0.6 degC per volt drift; settling about 0.65 s and about 0.10 degC RMS noise; SMBus at 100 kHz or less, address 0x5A; uncoated silicon lens

Block	Part	Datasheet facts that shape the design
Distance gate	VL53L1X	Class 1 eye-safe 940 nm laser, range to 4 m, inaccurate inside 4 cm, address 0x29
Gesture input	APDS-9960	Gesture range about 10 to 20 cm, address 0x39; a distributor reports the part is being phased out, so breakouts are for the prototype and a hover-sensor alternative is planned
NFC reader	PN532 module	Reads ISO 14443 tags up to about 70 mm, address 0x24; the bare chip is not recommended for new designs, so modules are used
Display	3.5-inch IPS SPI TFT (ST7796), no touch layer	About 300 cd/m2, 5 V, 95 mA backlight, 0.5 W; output only
Audio	MAX98357A amplifier and speaker	I2S, 3.2 W into 4 ohm at 5 V; spoken prompts
Indicators	WS2812 LEDs with a 74AHCT125 level shifter	5 V LEDs driven from 3.3 V logic
Clock	DS3231 RTC (optional)	Trusted timestamps when the site has no internet time
Sensor rail	Dedicated 3.0 V regulator	Powers the MLX90614 within its calibration tolerance

First spin is a carrier board with socketed modules, frozen only after breadboard bring-up gates pass. An optional MLX90640 thermal array is considered for a later phase.

Accuracy engineering

- The thermometer runs from its own 3.0 V rail on its own I2C bus with 3.0 V pull-ups, because a 3.3 V supply would add roughly 0.18 degC of error and its clamp diodes load a bus pulled above its supply.

- The firmware never polls the sensor continuously. It waits for a stable position, keeps the bus quiet, then reads a short burst, because continuous bus traffic can induce errors in narrow-field sensors.
- The sensor is thermally isolated from the amplifier, regulators and display backlight, and the enclosure aperture is wider than its 5 degree field of view.
- The lens is uncoated silicon with no optical filter, so the unit is sited away from direct sun and heaters, and a cleaning method is validated because contamination causes unspecified errors.
- The +/-0.2 degC figure is the sensor's specification in a defined window. The checkpoint's own accuracy is to be measured against reference instruments, not assumed.

Decision logic

Inputs are temperature, symptom answers and exposure answers. Thresholds live in a configuration file set by the Ministry of Health clinical protocol and can be updated without reflashing.

- **Green:** the person passes.
- **Amber:** triggers a re-measurement after a short rest, then escalation or clearance.
- **Red:** triggers referral per protocol.
- Every decision is logged with its inputs and the rule that fired.
- A sensor fault shows a manual-protocol message, so a failed unit never waves people through.

Identification and enrollment

Screening comes first and registration second, so an unknown person never blocks the line. An unknown NFC tag creates a provisional record keyed to the tag ID, and screening runs at once. Amber or red cases and monitored contacts go to an operator desk for enrollment with consent. Because the tag ID is the key, two checkpoints that see the same unknown tag offline end up with one record after they sync. Wristbands are single-use. Fingerprint and face recognition are deliberately not used: they need touch or create sensitive biometric data.

Infection control and enclosure

No touch input or keypad. The display is output only, behind its own sealed window. The enclosure is sealed, smooth, chlorine-resistant and rated IP65 or better with no open ports. A use counter locks the unit until a badge-confirmed cleaning. An operator barrier or PPE protocol applies.

Connectivity

The checkpoint sends to an on-site gateway over Wi-Fi (the gateway can be its own access point) using MQTT or HTTPS with TLS and per-device credentials. Every message is acknowledged, unsent records are buffered in flash, and screening never waits for the network. The gateway uplinks over LTE/GSM or the site's internet and stores and forwards when offline.

Electronic nose module (experimental adjunct)

The e-nose adds a second, non-thermal signal at the checkpoint, but it starts as a data-collection module that never changes a green, amber or red decision. No validated volatile-compound signature for Ebola is known so far, so the module makes no diagnostic claim. Its job is to record gas-sensor patterns next to temperature and symptom answers, so the team can test on labelled data whether they add anything. It becomes a decision input only after validation and Ministry of Health clinical approval.

Operating modes

1. **Shadow mode (default):** readings are recorded and linked to the patient ID, and nothing is shown to the person or the operator.
2. **Advisory mode (only after validation):** a validated pattern may prompt a re-measurement, never a stronger action on its own.
3. **Hard limits in every mode:** the e-nose can raise caution but can never turn an amber or red into a green, and a failed or purging module never blocks or alters screening.

Sampling design

- Standoff sampling only: a small inlet at face height beside the thermometer aperture draws ambient air at the person's head position into a sealed sensing chamber. Nothing touches the person.
- A replaceable hydrophobic filter sits in front of the chamber and another on the exhaust. Filters are changed on the use-counter cycle and handled as site waste under the health authority's protocol, because breath and droplets can load them.

- The chamber, fan and sensors sit inside the same sealed, wipeable enclosure design as the rest of the unit, with no open ports.
- An ambient baseline is captured before the person arrives, the exposure window runs while they answer the questions, and a purge follows. Window lengths are set during bring-up from measured sensor response.

Candidate sensors (to be confirmed against datasheets before the design is frozen)

Function	Candidate	Role
Metal-oxide gas array with humidity and temperature	Bosch BME688	Gas-resistance patterns under several heater profiles, with humidity compensation
VOC and NOx index	Sensirion SGP41	Second, independent volatile-compound signal
CO2	Sensirion SCD41	Marks exhaled breath and sets a dilution reference for the other channels
Air path	Small blower with hydrophobic filters	Moves air through the sealed chamber

Integration with the thermometer. The e-nose sensors sit on their own I2C bus and their own supply rail, separate from the MLX90614's 3.0 V rail and bus. The heated gas elements and the blower are thermally isolated from the infrared sensor. Firmware keeps the order fixed: the temperature burst runs first in a quiet bus, then gas sampling starts, so heaters and fan never overlap the thermometer's measurement window.

Confounders and drift

- Perfume, alcohol hand rub, smoke, food and strong odours are expected at every checkpoint, so samples with such signatures are flagged invalid and never scored.
- Chlorine-based disinfectant used for cleaning will saturate the sensors. After each badge-confirmed cleaning the unit runs a timed purge before the e-nose is marked ready.
- Humidity, ambient temperature and sensor ageing are tracked, with a reference purge and drift check at set intervals and a flag when a sensor needs replacing.

Data handling. Raw traces and extracted features are stored per screening against the pseudonymous patient ID, with signal-quality and confounder flags, and sent to the platform with the rest of the record. No e-nose result is shown to the person being screened.

E-nose validation gates

1. **Bench:** repeatability against known test gases, humidity and temperature sweeps, and drift over days.
2. **Confounders:** healthy volunteers with perfume, hand rub, chlorine cleaning, food and smoke, to define the invalid-sample rules.
3. **Shadow-mode collection:** recording alongside temperature and symptom answers at the pilot site, to measure the pattern in people who are not sick and the rate of invalid samples.
4. **Added value:** only with ethics approval and labelled outcomes from confirmed cases and negatives, test whether the e-nose adds anything beyond temperature and symptoms, against targets set with clinicians.

Limits. Confirmed Ebola cases are few and are handled only under laboratory biosafety, so Ebola-specific performance may not be measurable during this outbreak. In that case the module stays a research data-collection feature, and the checkpoint's decisions rest on temperature, symptoms and exposure answers alone.

7. Stage 2: GM Wearable

Purpose. Monitor contacts through the 21-day watch period and alert a health worker remotely when their vitals trend wrong, even when they are at home.

Hardware and the datasheet facts that shape it

Block	Part	Datasheet facts that shape the design
Controller	Fanstel BT840 family module (nRF52840)	Bluetooth 5 including long range, 1 MB flash, 256 kB RAM, VDD 1.7 to 3.6 V, -40 to 85 degC, FCC ID X8WBT840F; 16 castellated pins carry every v0.1 signal; approtect on the -p versions

Block	Part	Datasheet facts that shape the design
Skin temperature	MAX30205	+/-0.1 degC at 37 to 39 degC (sensor die), 2.7 to 3.3 V, about 600 uA, samples at least 10 s apart, address 0x48, 0 to 50 degC
Pulse and SpO2	MAX30102	1.7 to 2.0 V core, 3.1 to 5.0 V LED supply, address 0x57, proximity detection for on-wrist state
Motion	BMA400	Under 14.5 uA, activity recognition and wake-up, address 0x14
Power management	bq25120A	300 mA linear charger, buck 1.1 to 3.3 V in 100 mV steps, LDO, address 0x6A, 50 s I2C watchdog, ship mode
Wireless charging	bq51003 with a 19 mm WE-WPCC coil	Qi receiver for 2.5 W and below; resonant capacitors computed from the measured coil inductance in the final enclosure
Battery	Single-cell LiPo, protected	Sized from measured current draw

Power architecture

Qi coil to bq51003 (5 V) to the bq25120A, which charges the cell and provides a 2.9 V system rail (controller, MAX30205, BMA400) and a 1.8 V LDO rail (MAX30102 core). The MAX30102 LEDs run from the battery. At boot the controller drives the charger's CD pin high to enable its I2C interface, writes the rail settings, and must service the 50-second watchdog at least every 30 seconds, otherwise the rail falls back to 1.8 V.

Design note on the LED supply. The MAX30102 LED supply needs at least 3.1 V, while a LiPo cell falls toward 3.0 V near empty. The firmware therefore enforces a low-battery cutoff above 3.1 V for optical measurements, or the LED rail gets a small boost stage. Which one is used is settled by measured current draw during bring-up.

Measurement model

The wearable measures skin temperature at the wrist, which is not core temperature. It therefore reports trends against each person's own baseline and never claims a fever reading. It combines skin temperature, resting heart rate, SpO2, motion level and on-wrist state, each with a signal-quality flag. SpO2 is used as a trend input only, and its performance across skin tones is checked during validation.

Design target: one record a minute, stored locally for the full 21 days (about 30,240 records, roughly 0.48 MB at 16 bytes each, with external flash if needed).

Alert logic

Alerts trigger on sustained deviation from baseline, not single spikes, and on missed heartbeats, a removed device or a low battery. They escalate in tiers:

1. A status LED nudge to the wearer.
2. A health-worker call.
3. Dispatch.

The local alert works with no network at all.

Connectivity

Version 1 is phone-first. The wearable bonds to the wearer's phone app over encrypted BLE, and the app uploads to the platform with retries, falling back to SMS when there is no data connection. Records are kept until the platform acknowledges them, sequence numbers expose gaps and duplicates, and the phone corrects the wearable's clock at each connection. Standard Bluetooth health services are used where they fit. A later version adds an LTE-M/NB-IoT modem (nRF9151) for people without a usable phone, once local network bands are confirmed with the operators.

Security

Encrypted, bonded BLE; per-device keys held in the nRF52840 security subsystem; approtect to lock the debug port; over-the-air firmware updates; no names or contact details on the device.

Touch-safe design

Sealed shell, Qi charging with no open ports, replaceable wipeable strap, and one device per person unless fully decontaminated.

8. GM Platform

The platform is where checkpoint, wearable and laboratory data meet. It holds each person's record, shows a dashboard to health workers, supports decisions, and exchanges data with other health systems under consent.

Patient record

One pseudonymous ID links screenings, wearable readings, alerts and laboratory results. Names and contact details sit only in the platform's secured registry, never on the devices. Every reading carries a device ID, timestamp, sequence number and signal-quality flags, so bad readings can be told apart from real ones.

Dashboard

- A contact list colour-coded by status.
- A per-person timeline with data quality shown.
- A missed-check-in list.
- An alert queue with acknowledgment and escalation.
- A map of clusters.
- An audit log with role-based access.

Laboratory results

The platform accepts laboratory results as structured messages (HL7 FHIR) keyed to the patient ID and specimen number. A returning result updates the person's status, closes or raises alerts, and appears on the case timeline. This lets a negative result support the release of a contact from monitoring, and a positive result trigger the isolation and contact-tracing workflow, without any extra device in the field.

Decision support

Built in three layers, not jumped straight to AI:

1. **Rules set by a clinician,** such as sustained deviation from the person's own baseline, missed heartbeats and taken-off devices.
2. **Cohort analytics,** such as clusters by site and trends across checkpoints.
3. **Learned risk scoring,** only once labelled outcomes and clinical oversight exist.

The platform suggests and people decide, and every alert records the data that triggered it.

Data exchange

The platform exchanges data with other health systems using HL7 FHIR R4, with temperature, heart rate and SpO2 as observations and screening and laboratory results linked to the patient record. Kenya's Digital Health Agency publishes national FHIR implementation guides and operates a national Health Information Exchange; systems connecting to it must use FHIR, meet its security requirements, and exchange data only for authorized purposes. Aggregate reporting goes through the Kenya Health Information System (DHIS2), and the Digital Health Act 2023 requires interoperability and gives patients rights over their health data. The Agency's current implementation guides, including any outbreak or surveillance profile, are checked before the data mapping is built.

Exchange modes:

- A FHIR R4 interface that authorized systems can read from and write to using scoped access tokens.
- An outbound feed to the national Health Information Exchange once the platform has been certified and approved.
- Aggregate counts (screenings, alerts, confirmed cases) sent to the national reporting system.
- Inbound imports of confirmed laboratory results and demographics from facility and laboratory systems.
- Structured exports (FHIR bundles or CSV) for partners during the pilot.

Exchange controls: consent and purpose limits for every recipient, minimum data per recipient, de-identification for analytics and research, a full audit of every exchange, defined retention periods, and approval before any connection to a national system.

9. Infection control, privacy and security

- Touchless inputs; sealed, smooth, chlorine-resistant enclosures rated IP65 or better; no open ports.
- Operator barrier or PPE protocol, trained staff, and a biohazard waste procedure for the sampling that takes place alongside the system.

- Informed consent, encryption in transit and at rest, minimum data, and defined access and retention rules.
- Pseudonymous IDs on devices; names and contact details only in the secured registry; role-based access with a full audit log.
- No fingerprint or face recognition, to avoid touch and sensitive biometric data.
- Compliance with the Data Protection Act 2019 and the Digital Health Act 2023, with registration and impact assessment steps confirmed with the Office of the Data Protection Commissioner.

10. Validation plan

No accuracy claim is made until it has been measured. Validation runs one stage at a time, and each gate must pass before the next stage starts. Acceptance targets are set jointly with Ministry of Health clinicians.

Stage 1: GM Checkpoint

1. **Bench bring-up:** each sensor verified on the bus, the 3.0 V sensor rail and its tolerance measured, and thermal isolation from the backlight and amplifier checked.
2. **Thermometer accuracy:** readings compared with a reference blackbody source and a clinical reference thermometer across the working range and distance window.
3. **Environment:** performance checked across ambient temperature, near sunlight and near heat sources, to set siting rules.
4. **Decision logic:** the rules engine tested against scripted scenarios, including sensor faults, to show a failed unit never passes a person.
5. **Cleaning:** the disinfection method validated against lens and enclosure contamination effects.
6. **Field-like trial:** throughput and usability with volunteers, including gesture answers and spoken prompts in the languages used at the site.

The e-nose module has its own validation gates, set out under its subsection of Section 6, and does not hold up validation of the checkpoint's temperature function.

Stage 2: GM Wearable

1. **Bench bring-up:** power rails, watchdog servicing, charging and battery cutoff behaviour.

2. **Current draw:** measured over a full duty cycle to size the battery for 21 days of monitoring between charges.
3. **Sensor performance:** skin temperature, heart rate and SpO2 compared with reference instruments, including across skin tones and during movement.
4. **Alert path:** BLE relay, SMS fallback, missed-heartbeat alerts, offline buffering and sequence-gap detection tested with deliberately broken links.
5. **Durability:** sealing, disinfection cycles and strap wear.
6. **Volunteer wear trial:** baseline stability and false-alert rate over multiple days.

Platform

- FHIR conformance tested against the Digital Health Agency's implementation guides in a test environment.
- Load and failure testing of the alert queue and escalation paths.
- Security review, including access control and audit logging.

11. Regulatory and approval pathway

The items below are to be confirmed with each authority before pilot use.

- **Ministry of Health and the national command:** clinical protocol for thresholds and referral, and agreement to pilot within the response.
- **Pharmacy and Poisons Board:** classification and any approval or exemption route for the devices as medical devices.
- **Communications Authority of Kenya:** equipment approval for the radio components (Wi-Fi, Bluetooth, cellular).
- **Office of the Data Protection Commissioner:** registration and data protection impact assessment.
- **Digital Health Agency:** certification and approval before any connection to the national Health Information Exchange.
- **Research ethics and licensing:** ethics committee review and research licensing for any pilot involving human participants.

12. Pilot proposal

Aim. Show, at one site and within the national command's framework, that the GM Checkpoint can screen quickly without touch and that the GM Wearable and Platform can keep a group of consenting contacts continuously visible for the 21-day watch period.

Proposed scope.

- One or two checkpoints at a site chosen by the Ministry of Health (an airport, border post or hospital entrance).
- A small group of consenting contacts wearing GM Wearables alongside, not instead of, the existing daily follow-up.
- The GM Platform dashboard in the command centre, with laboratory results linked by patient ID where the laboratories agree.

What the pilot measures.

- Checkpoint accuracy against reference thermometers and screening time per person.
- Wearable data completeness, false-alert rate and time from trend change to health-worker acknowledgment.
- Alert acknowledgment and escalation times on the platform.
- Health-worker and participant feedback on usability, comfort and acceptability.

The pilot runs only after the validation gates above have passed and the approvals in Section 11 are in place.

13. Support requested

- **Clinical partnership:** a Ministry of Health or KEMRI clinical lead to set thresholds, review decision logic and co-sign the validation protocol.
- **Access to a pilot site and the command centre,** with agreement on how the platform's data relates to existing records.
- **Laboratory integration:** agreement with the National Virology Reference Laboratory and partners on a results feed keyed to the patient ID and specimen number.
- **Regulatory guidance:** introductions to the Pharmacy and Poisons Board, the Digital Health Agency and the ethics committee.
- **Funding and resources:** for prototype build, reference instruments for validation, test environments and pilot logistics.

- **Engineering and clinical review:** peer review of the design before the first build is frozen.

14. Risks and limits

- **Not a diagnostic.** The GM System flags people for assessment, testing and isolation. Forehead and wrist temperatures are not core temperature, and many people with early infection can look normal. Screening reduces risk; it does not remove it.
- **Sensor limits.** Infrared sensors are affected by sunlight, heat sources and contamination, and optical SpO2 sensors vary with skin tone and motion. Both are handled by siting rules, signal-quality flags and validation, not assumed away.
- **Component supply.** The APDS-9960 gesture sensor is reported as being phased out and the bare PN532 chip is not recommended for new designs. Modules are used for the prototype, and a hover-sensor alternative is planned.
- **Connectivity.** Version 1 relies on a phone for the wearable. People without a usable phone are covered by the planned cellular version, and in the meantime by the existing manual follow-up.
- **Privacy.** Health data about people under quarantine is sensitive. Pseudonymous IDs, consent, minimum data and audit mitigate this, and no external data sharing occurs without approval.
- **Not yet built.** Every performance statement in this document is a design intent until it has been validated on the bench and in the field.
- **E-nose limits.** No validated volatile signature for Ebola is known, gas sensors react to many everyday odours and to disinfectants, and Ebola-specific validation may not be possible during this outbreak. The module therefore runs in shadow mode and does not affect decisions until proven.

15. Roadmap

Phase	Work	Gate to move on
1. Bring-up	Breadboard the checkpoint and wearable sensors; build the platform skeleton and FHIR interface	Sensors and rails verified against datasheets

Phase	Work	Gate to move on
2. Checkpoint prototype	Carrier board with socketed modules, enclosure and decision logic; e-nose module added in shadow mode	Accuracy and cleaning validation passed
3. Wearable prototype	Sealed wrist unit with charging and BLE relay	Current draw, sensor and alert-path validation passed
4. Platform integration	Dashboard, alerts, laboratory results by patient ID, test-environment FHIR exchange	Conformance and security review passed
5. Approved pilot	One site, consenting participants, alongside existing follow-up	Approvals in place; results reviewed with the Ministry
6. Scale-up planning	Cellular wearable, hardware revision, wider deployment	Pilot results and clinical sign-off

Appendix: Key components at a glance

Stage	Key components
GM Checkpoint	ESP32-S3-WROOM-1-N16R2, MLX90614ESF-DCI, VL53L1X, APDS-9960, PN532 module, ST7796 display, MAX98357A, WS2812 LEDs with 74AHCT125, optional DS3231, dedicated 3.0 V sensor rail, experimental e-nose module (BME688, SGP41, SCD41 candidates)
GM Wearable	Fanstel BT840 (nRF52840), MAX30205, MAX30102, BMA400, bq25120A, bq51003 with 19 mm WE-WPCC coil, protected LiPo; later nRF9151 cellular
GM Platform	Site gateway (Wi-Fi in, LTE/GSM or internet out), phone app, backend and registry, dashboard, HL7 FHIR R4 interface, DHIS2 reporting, laboratory results feed